

ChE 451 – Renewable Energy Technologies –Fall 2025

Student Learning Outcome 1

Rubric for assessment on Midterm Hone Exam

ABET Outcome 2			
An ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.			
Performance Indicator (PI)	Level 3: Satisfactory	Level 2: Developing	Level 1: Unsatisfactory
1. Developing a design strategy	Develops a design strategy, decomposition of work into subtasks, development of a timetable	Uses a design strategy with guidance	No design strategy; haphazard approach
2. Applying engineering and/or scientific principles correctly to design practical processes	Applies engineering and/or scientific principles correctly to design practical processes	Includes only minor or cursory consideration of economic, safety, and environmental constraints	No application of engineering and/or scientific principles
3. Use of approaches	Suggests new approaches and improves on what has been done before	Can follow a previous example competently	Cannot design processes or individual pieces of equipment without significant amounts of help
4. Understanding how areas interrelate and demonstrates ability to integrate prior knowledge into a new problem	Understands how areas interrelate and demonstrates ability to integrate prior knowledge into a new problem	Can use prior knowledge to design individual pieces of equipment competently when guided to do so	Unable to relate prior knowledge to the design problem
5. Supporting design procedure with documentation and references	Supports design procedure with documentation and references	Minimal or incorrect use of computer tools and engineering resources	Design is done incompletely without the proper equations and without references
6. Developing a solution that includes realistic constraints (cultural, social, environmental, economic, health and safety factors)	Develops a solution that includes economic, safety, environmental and other realistic constraints	Design is done, but procedures and equations are not documented or referenced	No consideration of economics, safety, and environment
7. Measurement awareness of errors	Is aware of measurement error and does account for it statistically	Is aware of measurement error but does not account for it statistically or does so at a minimal level	Is unaware of measurement error

Student Outcome (2) Averages:

Performance Indicators: (1) 2.88 (2) 2.63 (6) 2.38 Overall Average: 2.63

PI (1) is the strongest score. The objective of this indicator is to evaluate the students' ability to develop a design strategy, and decomposition of work into subtasks for a techno-economic analysis for an off-grid solar+storage system to power a refugee camp. The students demonstrated excellent ability in estimating systems daily load, solar and battery sizing, and evaluating the overall cost of ownership over the lifetime of the project.

The next highest score is for PI (2) which corresponds to being able to apply engineering and/or scientific principles correctly to design practical processes. The students used basic solar energy principles to size the solar array, Li-ion battery capacity and maximum power demand over daily and annual basis. Some students struggled with differentiation between average daily energy need (average daily kWhr) which is covered by the solar and battery sizing versus the maximum power demand (i.e. maximum kW) which requires proper sizing of power electronics (i.e. inverters and converters).

Finally, the PI (6) is the lowest score. It corresponds to developing a solution that includes realistic constraints (cultural, social, environmental, economic, health and safety factors). Most students did not address the financing challenges and cost of capital for executing such humanitarian projects. In addition, the students did not appreciate the impact of cost of electricity for such displaced communities and their inability to afford such cost which may account for a large portion of their income. Finally, the students proposed alternative conventional power options, such as diesel generators, without addressing the environmental and health impacts of such sources and the challenges with accessing fuel supplies for such displaced communities. Including future course material to discuss socioeconomic challenges of various conventional and alternative energy solution should help improve students' comprehension of such challenges.